

Personal Summary

I am a second year PhD student at the University of Cambridge, investigating the potential impact of climate change on the distribution and incidence of bacterial meningitis. Before beginning my PhD I completed an MSc in Tropical Disease Biology at the Liverpool School of Tropical Medicine and worked as an Epidemiology and Information Analyst within the UK Health Security Agency Field Service (UKHSA). Whilst at UKHSA I worked on a wide variety of communicable disease outbreaks and undertook a secondment with the World Health Organisation (WHO) working as an epidemiologist on the MPox outbreak within Europe. In addition to my academic and professional work I have undertaken several voluntary roles within global health organisations, furthering my commitment to the field. I hope to pursue a career in infectious disease epidemiology, using my scientific training to support global public health programs, focusing on the integration of research into public health practice.

Education

University of Cambridge- Darwin College

Cambridge, UK

PhD in Veterinary Medicine

April 2024 - Present

Research Project: Charting Meningitis Outbreaks in Africa: Geospatial Analysis, Forecasting, and Optimizing Vaccine Control Strategies

Liverpool School of Tropical Medicine

Liverpool, UK

MSc in Tropical Disease Biology

September 2019 - September 2020

Research Project: Geospatial analysis of travel distance from health facilities in Malawi, in areas where acute African trypanosomiasis continues to be a public health issue

University of Kent

Canterbury, UK

BSc in Biomedical Science

September 2015 - June 2018

Dissertation Project: What is the correlation between waterborne parasites and flooding

Employment

University of Cambridge

Cambridge, UK

Research Assistant/PhD Student

January 2023 - Present

I am currently pursuing a PhD focused on understanding the potential impact of climate change on the distribution and incidence of bacterial meningitis. As part of this work, I have developed a predictive geographic model that estimates meningitis outbreak risk across Africa using temporal climate variables. Building on this, I am now using R-INLA to evaluate this relationship over space and time. Prior to my PhD, I conducted research on the epidemiology of pneumococcal meningitis in displaced populations within Africa, as part of a Research Assistant position at Cambridge which has since been published. .

UK Health Security Agency

Cambridge, UK

Epidemiology and Information Analyst

October 2020 - January 2023

Developed coding skills within R and R markdown through developing the care home COVID report to diverse stakeholder specifications. Piloted a syphilis survey for GUM clinics within Select Survey, providing an overview of patients' symptoms and treatment. Led Field Service response in a pressurised MRSA outbreak where I generated a case definition, analysed genomic data and wrote an outbreak report. Created the contact tracing data collection template for hospital and ambulance staff during Crimean Congo Haemorrhagic Fever incident, working independently to build an organised template within a time pressured and evolving event response. Conducted multivariable logistic regression to determine risk factors for late diagnosis of HIV within the East of England as part of a scientific research report.

World Health Organisation

Copenhagen, Denmark (UK Remote)

GOARN Epidemiologist

August 2022 - September 2022

Provided technical support within the European Mpox outbreak to subnational incident management teams, through the production of the WHO/ECDC bulletin and internal slide sets. This involved frequent updates to R coding, ensuring the addition of new analyses as requested by stakeholders as to adapt to data changes. Aided in the creation of the WHO/ECDC bulletin in Russian, working with translators to produce an Rmarkdown version of the bulletin using the Cyrillic alphabet. Monitored the implementation surveillance and response programmes through helping to design a survey examining response measures implemented in European countries affected by the ongoing multi-country MPox outbreak.

Academy of Nutritional Medicine

Cambridge, UK

Administrative Assistant

January 2019 - August 2019

Worked within the Academy of Nutritional Medicine (AONM) to provide educational and laboratory Test Services in the health field. Operated the helpline to provide customer support and liaise with patients via phone and emails exercising diplomacy and tact over confidential matters..

Voluntary Experience

Writer

DYYV Global

Cambridge, UK

March 2021 - January 2023

Wrote monthly articles for DYYV Global on a variety of global public health issues, with a focus on the epidemiology of tropical infectious diseases. As part of this I have conducted several interviews with global health charities regarding their disease elimination campaigns developing both my written and verbal communication skills.

Research Assistant

Liverpool School of Tropical Medicine

Liverpool, UK

December 2019

Volunteered to assist a PhD student in writing a systematic review as part of their thesis on the impact of HIV interventions aimed at enhancing socio-economic status through internal savings and lending schemes groups on mental health in low and middle income countries.

Youth Steering Group Member

VSO ICS

London, UK

May 2019- August 2019

Helped to organise and run the ICS Youth Employability Event which was attended by 60 former ICS volunteers. I worked within the Innovative Ideas and Speeches subgroup as well as the Workshop subgroup to help contact companies to come present at the event, co-moderate the discussion panel and aided in the formation of new ideas for the event.

Public Health and Epidemiology Intern

Lepra

Colchester, UK

January 2019- May 2019

Worked alongside an epidemiologist to research and deliver a project analysing active case finding in India between 1999-2005 which led to in-country leprosy elimination. Aided in the creation of several databases for donor support details. Planned and conducted an interview for International Women's Day with a leading health rights activist.

Global Alliance for Rabies Control

Volunteer

Cambridge, UK

July 2018- January 2019

Worked to put together a discussion document for internal staff at GARC regarding the WASH programme within GARC and any existing and potential coalitions/collaborations with similar organisations.

Programme Developer

VSO ICS

Mongla, Bangladesh

October 2018-December 2018

Undertook international field work in Mongla, Bangladesh as a volunteer within the sexual and reproductive health rights team. Through working with a multidisciplinary team we were able to carry out training sessions to local community educators and students sessions on topics including menstruation and skin bleaching. We also installed 2 sinks in local schools and ensured that several local girls could get access to vital medication.

Publications

Published

Cliff, M., Jahn, S., Fouda, A., Latt, A., Lingani, C., & Trotter, C. (Accepted for public peer review). Updating the meningitis belt: Associations between environmental factors and epidemic meningitis risk across Africa. *Wellcome Open Research*, 10:497. <https://wellcomeopenresearch.org/articles/10-497/v1>

Willgert, K., **Cliff, M.**, Meinke, S., Messina, D., Broom, D. M., Wood, J., Conlan, A. J. K., & Joshua, K. (2025). Transboundary and emerging diseases. *Transboundary and Emerging Diseases*, 2025-01. <https://doi.org/10.1155/tbed/6541298>

Cliff, M., Welaga, P., Mohammed, N., Ansah, P., Heyderman, R. S., Trotter, C., & Kwambana-Adams, B. (2024). Strategies for controlling pneumococcal disease and outbreaks during humanitarian emergencies. *Nature Medicine*, 30(6), 1515–1516.

In Preparation

Cliff, M., Diallo, K., Yapi, R., Tano, A., Jahn, S., Wainwright, C., & Trotter, C. How do climatic factors directly influence the incidence and risk of meningococcal meningitis across the African meningitis belt? A narrative literature review

Welaga, P., **Cliff, M.**, Mohammed, N., Ansah, P., Mwenda, J., Heyderman, R. S., Trotter, C., & Kwambana-Adams, B. Pneumococcal carriage and disease studies in the African Meningitis Belt between 2000–2020: A systematic review

Presentations

Poster Presenter

International Pathogenic Neisseria Conference

Florence, Italy

April 2025

Presented poster on association between environmental factors and epidemic meningitis risk across Africa.

Teaching

Seminar Leader/Supervisor

Cambridge, UK

University of Cambridge

September 2024- Present

Facilitated three seminar groups of 4–6 first-year Medical and Veterinary students, delivering instruction on the 'Foundations of Evidence-Based Practice' module. This module introduces core scientific principles underpinning modern clinical practice, including epidemiology and biostatistics.

References available upon request.